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**Mathem solvex
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 **Under the Guidance of
Amit Katiyar**



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AMIT KATIYAR

(MCA-JNU)



0512-3163515



9935985550



1. If $x = (2 + \sqrt{3})^{\frac{1}{3}} + (2 + \sqrt{3})^{-\frac{1}{3}}$ and $x^3 - 3x + k = 0$, then the value of k is:

(CUET-PG 2025) Algebra

- (a) -4 (b) 4
(c) $\sqrt{3}$ (d) $2\sqrt{3}$

2. if $\frac{1}{9!} + \frac{1}{10!} = \frac{x}{11!}$ then the value of x is:

(CUET-PG 2025) Equation Solving

- (a) 121 (b) 120
(c) 12 (d) 24

3. Match List-I with List-II

List-I	List-II
(A) X and Y are two sets such that $n(X) = 17$, $n(Y) = 23$, $n(X \cup Y) = 38$, then $n(X \cap Y)$ is	(i) 20
(B) If $n(X) = 28$, $n(Y) = 32$, $n(X \cap Y) = 10$, then $n(X \cup Y)$ is	(ii) 10
(C) If $n(X) = 10$, then $(7X)$ is	(iii) 50
(D) If $n(Y) = 20$, then $n\left(\frac{Y}{2}\right)$ is	(iv) 2

(CUET-PG 2025) Set Theory

Choose the correct answer from the options given below:

- (a) (A) - (iv), (B) - (iii), (C) - (ii), (D) - (i)
(b) (A) - (iv), (B) - (iii), (C) - (i), (D) - (ii)
(c) (A) - (iv), (B) - (i), (C) - (ii), (D) - (iii)
(d) (A) - (iv), (B) - (ii), (C) - (i), (D) - (iii)

4. From the given sets, which is an infinite set:

(CUET-PG 2025) Set Theory

- (a) $\{x: x \in \mathbb{N} \text{ and } (x-1)(x-2) = 0\}$
(b) $\{x: x \in \mathbb{N} \text{ and } x \text{ is prime number and less than } 199\}$
(c) $\{x: x \in \mathbb{N} \text{ and } x^5 - 1 = 0\}$
(d) $\{x: x \in \mathbb{N} \text{ and } x \text{ is odd}\}$

5. Match List-I with List-II

	List-I		List-II
(A)	If $\begin{bmatrix} \lambda - 1 & 0 \\ 0 & \lambda - 1 \end{bmatrix} = 0$, then λ is	(I)	0
(B)	If $\Delta = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$, then Δ is	(II)	1
(C)	If $A = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$, then $ A^{-1} $ is	(III)	-2
(D)	If $\begin{bmatrix} a+1 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 1 & 2 \end{bmatrix}$, then a is	(IV)	2

(CUET-PG 2025) Matrices & Determinants

(Eigen/Determinant/Inverse type matching)

Choose the **correct** answer from the options given below:

- (a) (A) - (II), (B) - (I), (C) - (IV), (D) - (III)
(b) (A) - (II), (B) - (I), (C) - (III), (D) - (IV)
(c) (A) - (II), (B) - (III), (C) - (I), (D) - (IV)
(d) (A) - (II), (B) - (IV), (C) - (I), (D) - (III)

6. If a, b and c in Geometric Progression and $a^{\frac{1}{x}} = b^{\frac{1}{y}} = c^{\frac{1}{z}}$ then, x, y, z are in _____.

(CUET-PG 2025) AP & GP

- (a) Arithmetic Progression
(b) Geometric Progression

(c) $\frac{2}{y} = \frac{1}{x} + \frac{1}{z}$

(d) $x = y + z$

7. Out of 5 consonants and 4 vowels, how many words of 3 consonants and 3 vowels can be made?

(CUET-PG 2025) P & C

- (a) 40 (b) 80
(c) 20 (d) 240

8. If $(x-1)$ is a factor of $2x^2 - 5x + k = 0$, then the value of k is:

(CUET-PG 2025) Polynomial Division

- (a) 2 (b) 5
(c) 3 (d) 4

9. The center and radius for the circle $x^2 + y^2 + 6x - 4y + 4 = 0$ respectively are:

(CUET-PG 2025) Circle

- (a) (2, 3) and 3
(b) (3, 2) and 8
(c) (2, -3) and 3
(d) (-3, 2) and 3

10. The length of major axis and coordinate of vertices for the ellipse $3x^2 + 2y^2 = 6$ respectively are:

(CUET-PG 2025) Ellipse

- (a) $2\sqrt{2}$, (0, $\pm\sqrt{3}$)
(b) $2\sqrt{3}$, (0, $\pm\sqrt{3}$)
(c) $2\sqrt{2}$, ($\pm\sqrt{3}$, 0)
(d) $2\sqrt{3}$, ($\pm\sqrt{3}$, 0)

11. If $x^2 = -16y$ is an equation of parabola then:

- (A) directrix is $y = 4$
(B) directrix is $x = 4$
(C) co-ordinates of focus are (0, -4)
(D) co-ordinates of focus are (-4, -0)
(E) length of latusrectum = 16

(CUET-PG 2025) Parabola

Choose the correct answer from the options given below:

- (a) (A) and (E) only
(b) (B), (C) and (E) only
(c) (A), (C) and (E) only
(d) (B), (D) and (E) only





12. Match List-I with List-II

List-I	List-II
(A) $\lim_{x \rightarrow 0} (1 + 2x)^{\frac{1}{x}}$	(I) e^6
(B) $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x$	(II) e^2
(C) $\lim_{x \rightarrow 0} (1 + 5x)^{\frac{1}{x}}$	(III) e
(D) $\lim_{x \rightarrow \infty} \left(1 + \frac{3}{x}\right)^{2x}$	(IV) e^5

(CUET-PG 2025) Integration

Choose the **correct** answer from the options given below:

- (a)(A) - (IV), (B) - (I), (C) - (III), (D) - (II)
 (b)(A) - (I), (B) - (IV), (C) - (III), (D) - (II)
 (c)(A) - (II), (B) - (III), (C) - (IV), (D) - (I)
 (d)(A) - (III), (B) - (II), (C) - (I), (D) - (IV)

13. The value of $\lim_{x \rightarrow \infty} \left(1 + \frac{2}{3x}\right)^x$ is:

(CUET-PG 2025) Limit

- (a) e (b) e^2
 (c) $e^{\frac{2}{3}}$ (d) $e^{\frac{1}{3}}$

14. If $f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$, then $f(x)$ is

(CUET-PG 2025) Continuity

- (a) continuous for all $x \in \mathbb{R}$
 (b) continuous at 0, 1 only
 (c) not continuous at 1
 (d) not continuous at 0

15. $\int \frac{2x+1}{x^2+x+2} dx$ is

(CUET-PG 2025) Definite integral

- (a) $\log(2x+1) + c$, where c is an arbitrary constant
 (b) $\log \frac{(2x+1)}{(x^2+x+2)} + c$, where c is an arbitrary constant
 (c) $\log(x^2+x+2) + c$, where c is an arbitrary constant
 (d) $\log\left(\frac{1}{2}\right) + c$, where c is an arbitrary constant

16. A fair coin is tossed three times. Let A be the event of getting exactly two heads and B be the event of getting at most two tails, then $P(A \cup B)$ is:

(CUET-PG 2025) Probabilities

- (a) $\frac{1}{2}$ (b) $\frac{3}{8}$
 (c) $\frac{1}{8}$ (d) $\frac{7}{8}$

17. Bag A contains 3 Red and 4 Black balls while Bag B contains 5 Red and 6 black balls. One ball is drawn at random from one of the bags and is found to be red. Then the probability that it was drawn from Bag B is

(CUET-PG 2025) Bayes Theorem

- (a) $\frac{35}{68}$ (b) $\frac{7}{38}$
 (c) $\frac{14}{37}$ (d) $\frac{34}{43}$

18. The points $(K, 2 - 2K)$, $(-K + 1, 2K)$ and $(-4 - k, 6 - 2K)$ are collinear if:

(CUET-PG 2025) Co-ordinate

- (A) $K = \frac{1}{2}$
 (B) $K = \frac{-1}{2}$
 (C) $K = \frac{3}{2}$
 (D) $K = -1$
 (E) $K = 1$

Choose the **correct** answer from the options given below:

- (a)(A) and (D) only (b)(A) and (E) only
 (c)(B) and (D) only (d)(D) only

19. In an examination, a student scores 4 marks for every correct answer and loses 1 mark for every wrong answer. If she/he attempts all 60 questions and secures 130 marks, the number of questions she/he attempts wrongly, are?

(CUET-PG 2025) Number System

- (a) 38 (b) 22
 (c) 21 (d) 37

ANSWER KEY

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
(a)	(a)	(a)	(d)	(a)	(a)	(a)	(c)	(d)	(b)
11.	12.	13.	14.	15.	16.	17.	18.	19.	
(c)	(c)	(c)	(a)	(c)	(d)	(a)	(a)	(b)	

